The bacteriological diagnosis of enteric fever consists of the following methods, which are:

- Isolation of the bacilli
- Demonstration of antibodies

Isolation of the bacilli

The typhoid bacilli are isolatedfrom the following clinical specimens which are tabulated (Table 7.14).

Demonstration of Antibodies: Slide – agglutination: The isolate is identified by slide agglutination with 'O' and 'H' antisera.

Widal Test: It is an agglutination test for detection of agglutinins 'H' and 'O' in patients with enteric fever. Salmonella antibodies start appearing in the serum at the end of 1st week and rise sharply during the 3rd week of enteric fever.

7.9.6 Prophylaxis

Various types of vaccine and their doses are given in Table 7.15.

Table 7.15: Various types of vaccine and their doses.

Vaccine	Doses
TAB – Vaccine	2 doses of 0.5 ml at an interval of 4–6 weeks
Typhoral	3 doses on alternate days. It gives 65–96% protection for 3–5 years and is safe
typhim – Vi	A single dose of 25µg

Table 7.14: Isolation method of typhoid bacilli from various clinical specimens.

Specimen culture	Isolation methods
Blood culture	5–10 ml of blood is collected and inoculated into blood culture bottle containing taurocholate broth or bile broth. After overnight incubation at 37°C the taurocholate broth is subculture on MacConkey agar. Pale colonies (NLF) appear on MacConkey which is used for motility and biochemical reactions.
Clot Culture (An alternative method to blood culture)	5 ml of blood is collected into a sterile test tube and allowed to clot. The clot is broken up with a sterile glass rod and added to bile broth containing streptokinase, which digests the clot and there by the bacilli are released from the clot. Then it is subcultured on MacConkey agar.
Faeces culture	Faeces sample are inoculated directly on MacConkey's agar, DCA or SS agar. The plates are incubated at 37°C for 24 hours, then characteristic colonies are observed which is confirmed by gram staining.
Urine culture	Urine samples are centrifuged, and the deposit is inoculated into enrichment media and then on selective media.

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